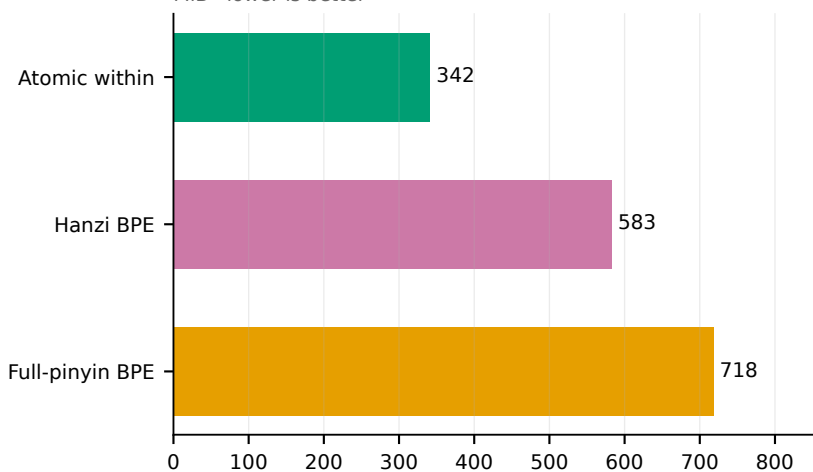


Atomic-within efficiency relative to standard representation baselines

Atomic within Hanzi BPE Full-pinyin BPE

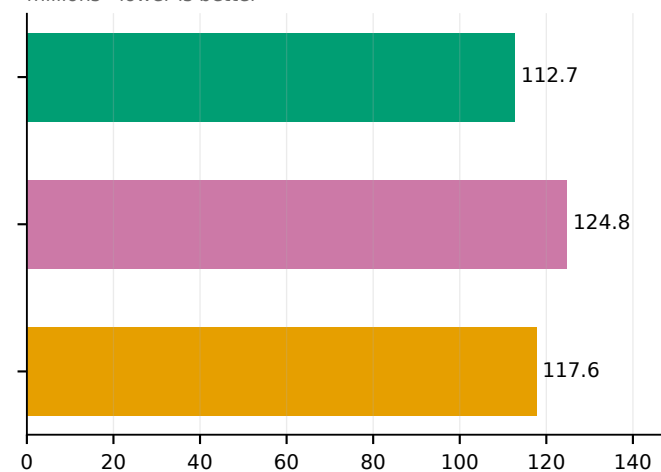
A Processed text storage

MiB · lower is better



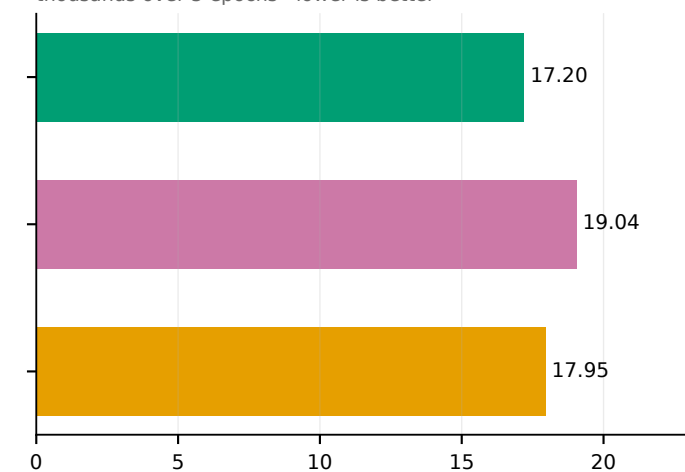
B Tokens per epoch

millions · lower is better



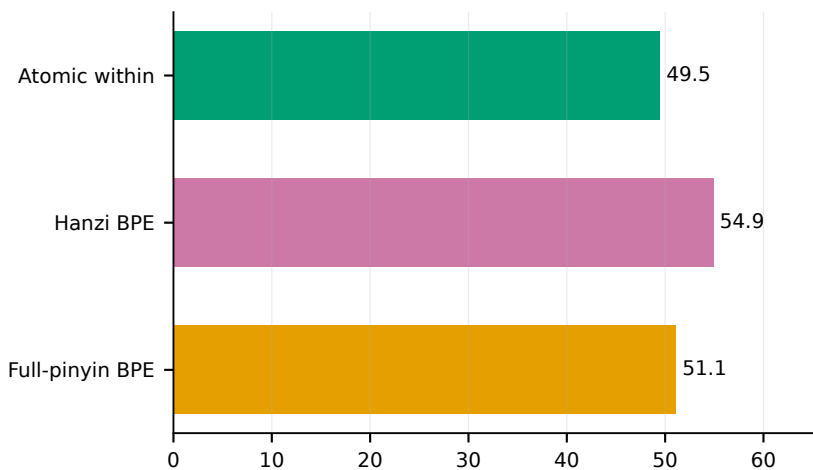
C Optimizer steps

thousands over 5 epochs · lower is better



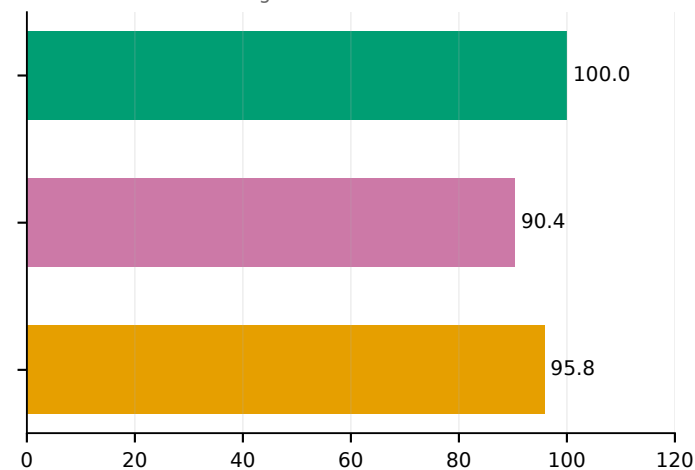
D Observed A100 time

minutes · lower is better



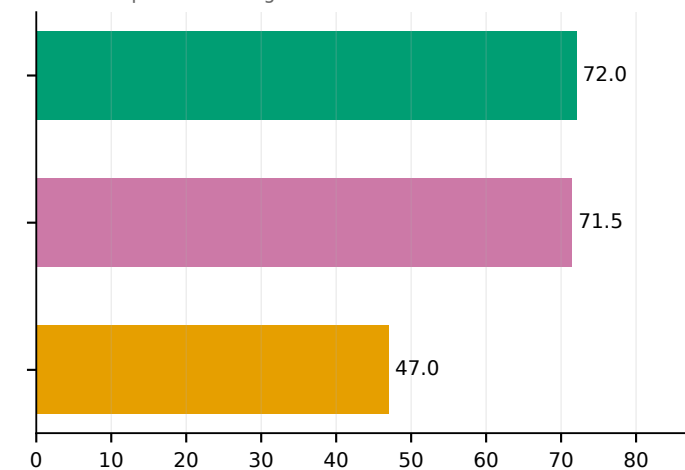
E Fixed-context coverage

Atomic within = 100 · higher is better



F Composite performance

all-model percentile · higher is better



Versus Hanzi BPE: 41.4% less text storage · 9.6% fewer tokens · 9.7% fewer steps · 9.9% shorter observed run · 10.7% more source coverage per context

Versus full-pinyin BPE: 52.4% less text storage · 4.2% fewer tokens · 4.2% fewer steps · 3.2% shorter observed run · 4.4% more source coverage per context

Same source documents, 97.7M-parameter GPT-2, 16k vocabulary, 512-token context, and five epochs. Training time is one observed run per model, not a replicated hardware benchmark.